Chapter 1
Economics

Preview

- The aim of this chapter is to introduce the reader to economics.
- The discipline of economics is defined and the objectives explained.
- Six building blocks of economic theory are reviewed.
- We believe that a good economic analysis of infection control should rest on strong foundations.

1.1 A Broad View of Economics

An economic system – big or small – must select and then structure its activities. It might be a national economy or a medium size hospital, but three key questions are the same:

(i) *What* should be done?
(ii) *How* should it be done?
(iii) *Who* will enjoy the benefits of the chosen activities?

Economics is concerned with answering these questions.

A national economy such as the United States or the United Kingdom is the sum of its resources, which fall in two groups, capital and labor. Capital resources include land, fossil fuels, buildings, computers, heavy machinery, brand names and reputation. Labor is the time individuals spend working, whether waiting on tables, cleaning or clerical work, or in more structured professions such as accountancy, architecture, or medicine. How capital and labor resources are used define an economy. This is important because resources are finite, yet there are many different ways to use them. Getting it wrong and making bad choices about how to use scarce resources can lead to great hardship and poverty. Successful economies, where resources have been allocated with some efficiency, deliver a good standard of living for the majority of the population.

A medium size hospital is also the sum of its resources. It comprises capital resources, from diesel generators to surgical robots, and many different types of labor, from cooks to cardiologists. Those who manage hospitals have to think long

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and hard about how they use resources because like those of national economies, resources are finite and can be used in many different ways. Making bad choices about how to allocate a hospital's resources between competing activities can cause worse health outcomes for patients.

We have decided that cracking nuts is a useful activity that we wish to devote scarce resources to. We have a budget of $20 (our scarce resources) that can be used to acquire a 'hammer' or a 'nut cracker' in order to complete the task (See Figure 1). Using our scarce resources to buy the nut cracker provides more benefits than if we had chosen the hammer. Making good choices about how to allocate scarce resources is central to economics. The best decision is one that maximizes benefit from the resources available. As long as you understand this, you are starting to think like an economist.

Economists work with politicians, business people, lawyers, and academics to allocate resources that make-up an economic system, to obtain the best value or benefits for society. The aim of economics is to make the maximum number of people as happy as possible, given finite resources. To achieve this requires careful consideration of those three key questions:

(i) *What* should be done?
(ii) *How* should it be done?
(iii) *Who* will enjoy the benefits of the chosen activities?

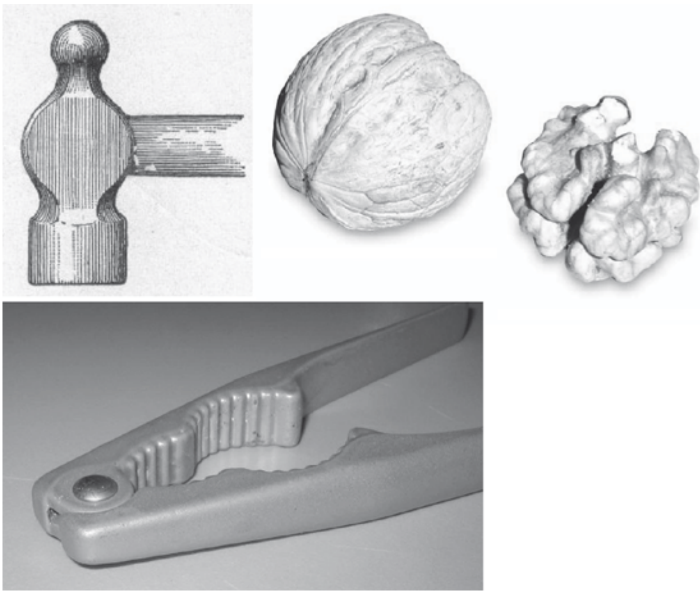


Fig. 1 Two ways to crack a nut, one better than the other

Economics is used in two ways. "*Positive Economics*" explains *what actually happens* in society. It answers "what is" type questions, such as, "What is the effect of allocating scarce resources in order to host the Olympic Games?," "*Normative Economics,*" on the other hand, is used to identify *what should happen*. It answers questions about "what ought to be," such as whether the Olympic Games should be hosted. Is it more valuable to society than some other use of scarce resources, such as poverty relief or improved access to healthcare? Normative questions stimulate debate because they require value judgments to be made. Positive decisions are less contentious because they are based on facts. Both forms of economics are concerned with the study of how resources are allocated within an economic system. This book is concerned with how resources are used for the purpose of infection control. This implies that both positive and normative questions will be addressed. We need to gather facts about what infection-control might cost and what benefits it may provide, yet we also need to make judgments on how valuable those outcomes are compared with other ways of using scarce healthcare resources. This book is a mixture of positive and normative economics.

Some economic concepts that will be useful throughout this book are described in the next section. These are Scarcity, Opportunity Cost, Efficiency, Competitive Markets, Market Failure, and Economic Appraisal. Our objectives are to define these terms using nontechnical language, bring them together like building blocks and encourage you to think like an economist. Economics is useful if you want to show why something should change. This book is about preventing healthcare-acquired-infections and we aim to provide readers with enough economics to make rational and well constructed arguments about why something should change with the current allocation of infection-control resources. You can make your economic arguments to a middle manager in your hospital or the chief executive, local or regional health planners, or state or national politicians. The principles and arguments remain the same and are quite simple. You may wish to argue for more resources to be devoted to infection-control and away from some other activity, or for existing infection-control resources to be used in another way. We offer you a set of tools and a way of thinking that can improve how decisions are made and, ultimately, how scarce resources are used for infection control.

1.2 The Building Blocks of Economics

1.2.1 The Concept of Scarcity

Scarcity is at the core of economics. Not all of societies' objectives can be achieved with the available resources. Although it might be desirable to improve education and healthcare services, this would require cuts to some other part of the economy, such as law enforcement or road maintenance. Every household might like to have a new car and three foreign holidays a year, but this might require cuts to mortgage payments or retirement savings. Scarcity arises because humans have a voracious desire to be successful. Some want material goods, big houses, and sports cars; others desire recognition in the community and want to help others; while some may just desire a simple but comfortable life in the countryside. However, the capital and labor resources required to make all these things happen are finite.

Hadley Cantril [14] collected some data that illustrate the range of human wants. The people interviewed lived in India or the United States, and here are some extracts:

India – 35-year-old man, illiterate, agricultural laborer.

> *"I want a son and a piece of land… I would like to construct a house of my own and have a cow for milk and ghee. I would also like to buy some better clothing for my wife. If I could do this then I would be happy."*

India – 45-year-old housewife.

> *"I should like to have a water tap and a water supply in my house. It would also be nice to have electricity. My husband's wages must be increased if our children are to get an education and our daughter is to be married."*

United States – 34-year-old laboratory technician.

> *"I would like a reasonable enough income to maintain a house, have a new car, have a boat, and send my four children to private schools."*

United States – 28-year-old Lawyer.

> *"Materially speaking, I would like to provide my family with an income to allow them to live well - to have the proper recreation, to go camping, to have music and dancing lessons for the children, and to have family trips. I wish we could belong to a country club and do more entertaining."*

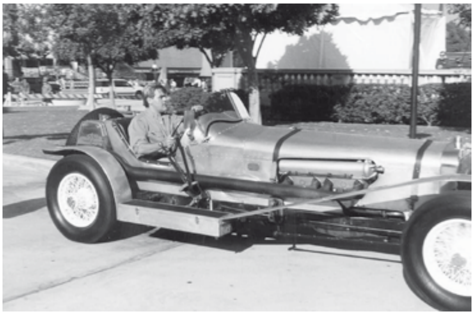
These individuals all strive for more. They reveal material and social aspirations for themselves and their families (i.e., a cow, tap water, a boat, or dancing lessons); the difference is that some have more resources than others and so prefer different things. Most Americans take running water, electricity, and a house for granted, yet these extracts reveal that this is not the case in India. The argument we make is that humans always want more of something, regardless of the circumstances.

John Maynard Keynes is the father of macroeconomics and made a prediction in the 1930s. He suggested increases in wealth would lead to a situation where individuals would have all their needs (demands) met and so would cease to strive for more of anything, instead devoting further energies to "noneconomic purposes" [15]. The examples of Imelda Marcos and her enthusiasm for shoes and Jay Leno and his passion for motoring, suggest otherwise. Human wants will never be fully satisfied, and because resources are finite, choices will always have to be made about how they are used.

The more productively resources are used the more goods and services are available. The result is that citizens of some countries have more material possessions than others. The American and Indian economies are both rich in resources, yet the American economy extracts more value from available resources when compared

Jay Leno is a car enthusiast with an impressive collection of modern and vintage vehicles. The collection is always growing, even though he can only drive one at a time.

In 1986 Imelda Marcos and her husband fled the Philippines after a revolution against a corrupt regime. She left behind mink coats, 508 gowns, 888 handbags and 1060 pairs of shoes.



with the Indian economy. This difference will diminish if the Indian economy continues to grow rapidly. Decisions are made everyday in both countries about what can and what cannot be done with scarce resources, and this is because of scarcity. Scarcity is the core of economics.

Paul Samuelson wrote this definition of economics [16],

> *"economics is the study of how men and society end up choosing, with or without the use of money, to employ scarce productive resources that could have alternative uses, to produce various commodities and distribute them for consumption, now and in the future, among various groups in society"*

He goes on to describe the three challenges for any economy.
Challenge One: *What* commodities shall be produced and in what quantities?
Challenge Two: *How* should they be produced?
Challenge Three: *Who* will enjoy the benefits of the commodities?
The process of answering these questions will give rise to an allocation of resources. It is important to answer these questions as appropriately as possible. Because of scarcity, when a choice is made, an opportunity to do something else is lost and this is called an opportunity cost.

1.2.2 The Concept of Opportunity Cost

We cannot do everything we want (think about scarcity), and a decision to use resources in one way causes a loss elsewhere in the economy. The loss arises because resources are no longer available for an alternative use. If a hospital manager decides to screen every new admission for methicillin-resistant *Staphylococcus aureus* (MRSA), then some resources would be used up. These include *labor resources*, such as the time of doctors, nurses, microbiologists, and infection-control professionals, and *capital resources*, such as isolation beds, gowns, gloves, microbiology equipment, and other consumables. Without the screening program, these resources could have been used to treat 500 new patients. The benefits from treating these extra patients represent the opportunity costs of the MRSA screening program. How these opportunity costs are valued depend on the perspective of the analyst. The hospital manager would look at the financial revenues foregone from not admitting the 500 new patients. A public health professional might estimate the years of life saved from diagnosing and treating the symptoms of the 500 patients. These two estimates are mutually exclusive. We cannot count the lost financial revenues to the hospital and the years of life lost, that would be double-counting. We could attempt to attach a dollar figure to the years of life lost and represent dollar estimates of opportunity costs this way. Economists tend to think about broader social objectives when making suggestions about how resources should be used and would therefore prefer the second approach to estimating opportunity costs.

There are situations where the opportunity cost of a decision is zero, or close to it. If there were no new patients waiting to be admitted to the hospital, staff were protected by long-term employment contracts and so could not be released, and if there was a stock of consumables that had to be used before an imminent expiry date, then the opportunity costs of the MRSA screening program would be zero. The reason is that the resources could not be used in another way and so no benefits have been foregone by deciding to screen for MRSA. If there were no patients waiting to be admitted but the resources could be turned into cash by firing staff and selling the consumables to another hospital, then the cash generated would represent the opportunity cost of the MRSA screening program.

The United States administration of the early 1960s made a decision to send man to the moon and return him safely to the earth. This project used billions of dollars worth of scarce resources and was built on the vision of Werner Von Braun, an active member of the "Society for Spaceship Travel" in 1930s Germany. An economist would ask what else could have been done with the resources. On the brink of the launch of Apollo 11, powered by the Saturn rocket illustrated in Fig. 2, Reverend Ralph Abernathy led a march of the Poor People's Campaign to Cape Canaveral to make exactly this point. He spoke with Thomas Paine, the NASA administrator at the time, and suggested the opportunity costs of the Apollo program were great, highlighting the poverty and suffering in American society. He argued poverty and suffering could be reduced with a reallocation of resources away from space exploration and toward interventions that reduce the causes and consequences of poverty. Whether Reverend Abernathy had a good economic argument depends on the value of the Apollo program relative to a poverty alleviation program or any other competing uses of scarce resources.

Some made-up data that describe the benefits of the Space program and the competing alternative ways of using scarce resources are included in Table 1.

Interpreting these data is quite simple. We choose the alternative that provides the greatest net benefit. Because we are constrained by scarcity, we only have enough resources to implement one program, and all programs use the same

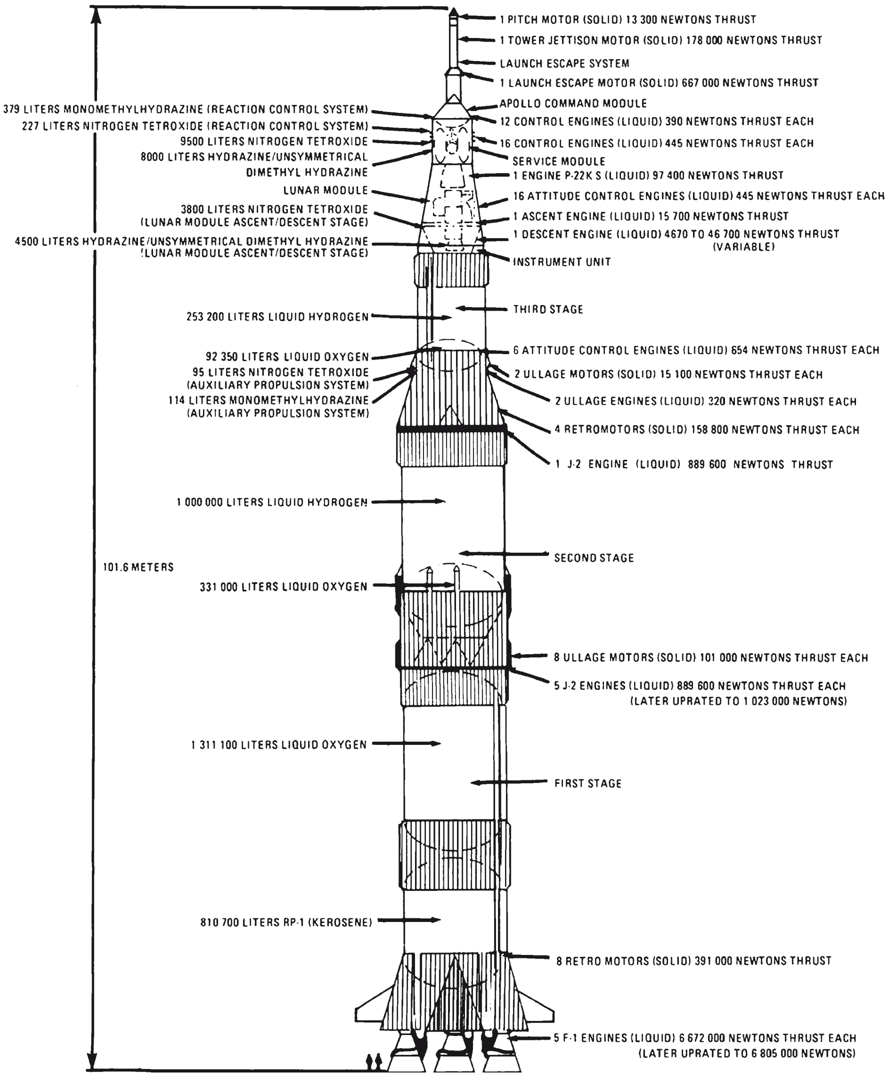


Fig. 2 The Saturn rocket used for the space program that put man on the moon

amount of resources. The decision to choose one program implies a loss, as we have rejected the others. The value of this loss is the opportunity cost. The decision to choose the space program implies a loss of benefits from poverty alleviation and these are valued at $120 billion. This is the opportunity cost of the space program. As the decision to pursue the Space program provides benefits worth $150 billion then we are better off by $30 billion.

Table 1 An example of opportunity cost

Options	Description of the economic benefits	Dollar valuation of economic benefits[a]
Space Program	Scientific advances Commercial spin offs Cold war supremacy Extending frontiers	$150 billion
Poverty Alleviation	Improved health Reduced crime Better workers Increased tax revenue	$120 billion
Tax cuts for the wealthy	Increased savings Increased spending Increased luxury car sales Increased overseas travel	$60 billion

[a] We assume that estimating the dollar values for economic benefits is easy. In reality, it is difficult and requires normative judgments to be made. Economists and policy makers argue long and hard about how to value things that are not traded in competitive markets (i.e., easy to acquire from shops or other suppliers), and these debates have shaped how economic analyses are structured and undertaken

The complete decision is summarized like this:

Space program: Benefits of $150 bn less opportunity costs of $120 bn = positive $30bn

Poverty Alleviation: Benefits of $120 bn less opportunity costs of $150 bn = negative $30bn

Tax cuts for the wealthy: Benefits of $60 bn less opportunity costs of $150 bn = negative $90bn

The space program delivers positive net benefits of $30 billion and the other two options necessarily incur net losses. Take a little time to make sure you understand this. This conclusion leads to a simple decision-rule that requires only those activities that provide the maximum difference between benefit and cost to be chosen, given scarce resources. This is equivalent to making decisions that maximize net benefit. We prefer to allocate scarce resources to programs that maximize net benefit because all members of society can be made better off. Those who enjoy the economic benefits can cover all the costs, and there is enough left over to compensate everyone else. If we choose a program that leads to losses, then there is no way to finance the program without leaving some people worse off. If this rule is applied to all possible decisions in society, then the maximum amount of benefit will be generated.

Table 2 An example of opportunity cost relevant to infection control

Competing decisions	Description of the economic benefits	Dollar valuation of economic benefits[a]
MRSA screening	Reduced rate of infection Shorter length of stay Reduced mortality	$100,000
Surgical robot	Better surgical outcomes Shorter procedure duration Reduced mortality	$60,000
Upgrade to the hospital IT system	Improved user interface More accurate billing Link to primary care records	$48,000

[a]Valuing the economic benefits from healthcare is also difficult. Lives may be at risk and so the consequences of decisions are potentially devastating to those involved. The methods for valuing a human life are important for health economics and these are described at the end of this chapter

Now consider an example from the hospital setting. The management board has resources available to implement one of three programs: MRSA screening, the purchase of a surgical robot, or an upgrade to the hospital information technology (IT) system. Data on the benefits of these programs are summarized in Table 2.

The application of the decision rule suggests net benefits of the "MRSA screening" program are $100,000 less $60,000 opportunity costs = positive $40,000 and so choosing one of the other alternatives will lead to net losses (i.e., the robot has $60,000 benefit less $100,000 opportunity costs which equals losses of $40,000, and the IT system has $48,000 benefit less $100,000 opportunity costs which equals losses of $52,000). Opportunity costs have been calculated by identifying the value of the benefits foregone with each decision. With this information we would always choose MRSA screening. The surgeon and users of the IT system who have incurred a loss will have to be placated, but decisions are made on behalf of the whole community. The adoption of the space program and the MRSA screening program enhance efficiency by the allocation of scarce resources.

1.2.3 The Concept of Efficiency

Economics is about scarcity, opportunity cost, and using resources such that benefits to society are maximized. Improvements in an economic system need to be measured, be it a whole country or a single hospital, and for this we use a yardstick called economic efficiency. Economists are nearly always interested in improving economic efficiency. Economic efficiency is about choosing the optimal use (i.e., allocation) of scarce resources. An economic system is said to be economically efficient when scarce resources are used to produce the maximum amount of goods and services that individuals want. Two conditions must be met before economic efficiency can be achieved. These are "allocative" and "productive" efficiency. The concepts are illustrated by Fig. 3.

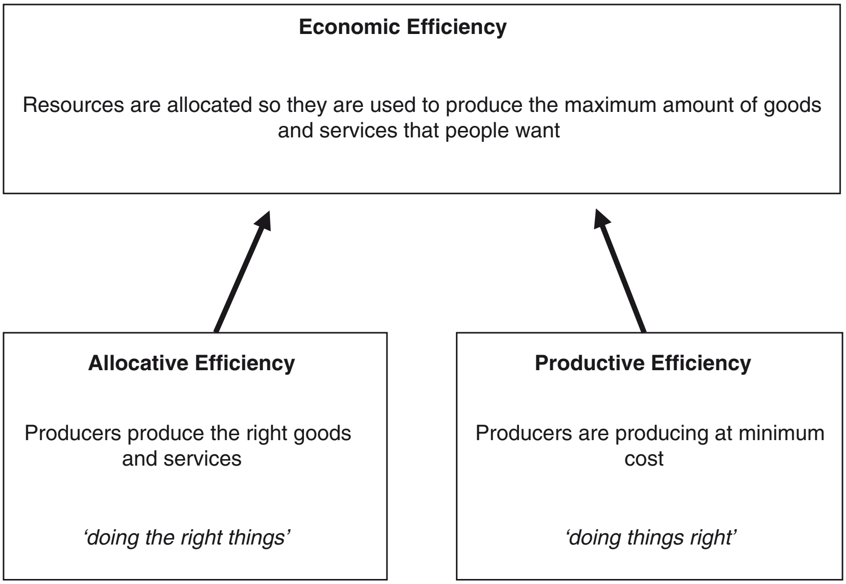


Fig. 3 Three types of efficiency

Allocative efficiency is when producers choose to produce the right goods and services, they are "*doing the right things.*" We used hypothetical data to show that the Space Program and MRSA screening were the right things to do. Productive efficiency is when producers make these goods at minimum cost; they are "*doing things right.*" Those who organized the Space program and the MRSA screening must pursue these activities at minimum costs and waste nothing. For example, MRSA screening could be done at a routine preadmission clinic, or a separate consultation could be organized and a dedicated MRSA nurse employed to perform the swabs. The former is likely to be more productively efficient (cheaper) than the latter.

Vilfredo Pareto was a well known philosopher and economist who died in 1923. His interpretation of economic efficiency was an allocation of resources where no individual can be made better off without another being made worse off, from any further reallocation. A balance has been reached where scarce resources are being used to produce the maximum amount of goods and services that people want. To achieve this outcome, decisions like those to adopt the space or MRSA screening programs would have to be made until no further gains were possible from scarce resources. The winners from the space program would have received net benefits worth $30 billion and according to Vilfredo Pareto, they must compensate everybody else. The $30 billion profit is the reason we chose the "Space program" in the first place. The other programs would have left us worse off. The winners from the MRSA screening program enjoyed benefits worth $40,000, and they would have to compensate everyone else who missed out. Again, the $40,000 profit is why we chose MRSA screening.

Making these compensations in reality is difficult and so economists suggest the compensations do not actually have to happen, but instead only the possibility for compensation has to exist. This rule, known as the Kaldor-Hicks criterion, is not perfect, but it does add flexibility to decision making and oils the cogs of economic efficiency. The rule allows decisions to be made where some individuals are winners and others are losers; furthermore, the winners do not actually have to compensate the losers, they just have to be able to make the compensations.

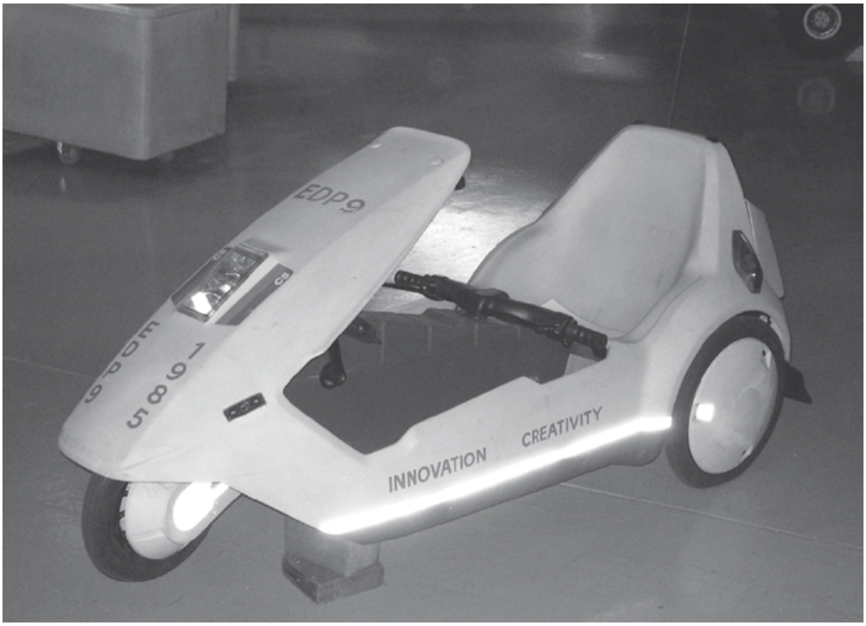
There are two mechanisms in an economy that influence resource allocation under scarcity. The first is the use of “Competitive Markets” and the second is the use of “Economic Appraisal.” Because these two mechanisms inform resource allocation, they impact on efficiency. The workings of the market and the process of economic appraisal are the subjects for the remaining sections of this chapter.

1.2.4 The Concept of Competitive Markets

A market can be used to allocate scarce resources and so address the three key questions for an economic system. Markets encourage infinite cost-benefit calculations amongst all those who participate, just the same as were done to choose the space program and the MRSA screening programs. A market is a physical or virtual environment where producers and consumers interact. A fruit and vegetable market held in the local park every Saturday morning is an example of a physical market, and the global market for iron ore is a virtual market as you do not have to be physically present to participate. The role of a market is to allow producers and consumers to exchange goods and services. It is driven by the rule of maximizing net benefits and has the desirable outcome of allocating scarce resources efficiently. Consumers demand goods and services in a market, and every consumer knows exactly how valuable the items are. Producers learn about what consumers want (i.e., demand), and then meet those demands by allocating scarce resources to productive processes. Consumers are “sovereigns” as they ultimately decide how producers use scarce resources. The market is an environment that disciplines producers into making things that enhance economic efficiency. A producer who allocates scarce resources to manufacturing goods and services that consumers do not value the most will be fail in the market.

The Sinclair C5 illustrated here suffered the wrath of the market. Although some might say this product was ahead of its time, the consumers of 1985 were not impressed with this plastic, battery-powered single-seater. A journalist who tested this vehicle concluded:

> *“I would not want to drive a C5 in any traffic at all. My head was on a level with the top of a juggernaut’s tyres, the exhaust fumes blasted into my face. Even with the minuscule front and rear lights on, I could not feel confident that a lorry driver so high above the ground would see me. Small wonder that one of the accessories listed in the C5 brochure is a high and bright-red reflecting mast.”*



Only 12,000 were produced and very few were demanded by consumers. The C5 was a commercial disaster. Consumers exerted their sovereignty, signaled that the C5 was not going to increase economic efficiency and ensured no more resources were allocated to its production.

In contrast, consumers loved the Volkswagen Beetle. Demand was strong for this cheap, reliable, and quirky car from 1938 to 2003, when production in Mexico finally ended. The less popular but more beautiful VW Karmann Ghia was a success too. Consumers sent signals to producers about what to make and how to make it. This allocation of resources (away from C5s and toward Volkswagens) was made by the "*invisible hand*" of the market [17].

Adam Smith first used this phrase – the invisible hand – in 1776 to argue that the goods and services of most benefit to society will, naturally, also be those that are most attractive for producers. The force behind the invisible hand is the price system, which is at the heart of a competitive market. The data included in Fig. 4 demonstrate how the price system works. If the price of a physical item or service is $5, then consumers in the market only demand one item. They seek alternate ways of obtaining benefits from their scarce resources in other markets. Yet, at $5, producers want to make five units because the price is high and they can make good money. This market is out of balance as there is a surplus of four unsold items. Consumers are busy spending their money in other markets where they get better value for money.

When the price is only $1, consumers want five items because they are now represent good value for money, yet producers only want to produce one item because of the small profit at the low price. The market is again out of balance, and there is a shortage of four items. Producers are busy using their resources to produce goods and services that can be sold in more profitable markets. Because of this shortage, consumers now bid against each other for the item and the price rises. This will stimulate producers to allocate resources toward production, as they see profits at higher prices. When the price reaches $3, both parties are satisfied and the market has reached a balance. The price tells consumers how to spend their money, and the producers follow this lead by making what the consumers want in exactly the way they want it. If every consumer is allowed to choose what they buy and producers are allowed to choose what to produce, then the market will find the balance. This is an allocation of resources that produces the maximum amount of goods and services that individuals want and represents a state of economic efficiency.

WHEN THE PRICE IS:	THIS QUANTITY IS:			THERE IS A MISMATCH OF:
	wanted	*offered*	*traded*	
$5	I	IIIII	I	surplus of IIII
$4	II	IIII	II	surplus of II
$3	III	III	III	zero: market in balance
$2	IIII	II	II	shortage of II
$1	IIIII	I	I	shortage of IIII

Fig. 4 How the price system works

Joseph E. Stiglitz [18] summarizes these ideas:

> *"If there is some commodity or service that individuals value but that is not currently being produced, then they will be willing to pay something for it. Entrepreneurs, in their search for profits, are always looking for such opportunities. If the value of a certain commodity to a consumer exceeds the cost of production, there is a potential for profit, and an entrepreneur will produce that commodity. Similarly, if there is a cheaper way of producing a commodity than that which is presently employed, an entrepreneur who discovers this cheaper method will be able to undercut competing firms and make a profit. The search for profits on the part of entrepreneurs is thus a search for more efficient ways of production and for new commodities that better serve the needs of consumers."* (pp. 56–57)

Perfectly competitive markets seem a very smart way to allocate scarce resources. Economic efficiency is achieved, net benefits maximized in society and scarce resources are used efficiently. The first two questions of economics, (i) *what* should be done and (ii) *how* should it be done, have been addressed. The third question, (iii) *who* will enjoy the benefits of the activities, is also addressed because consumers express their preferences for certain goods and services in markets.

In a perfect world, the initial distribution of wealth is fair with an extra dollar of benefit valued equally by any member of society, for example, a poor American living in rural Kansas would value another dollar exactly the same as would the chief executive of a large company building fuel tanks for the Saturn rocket. In the real world (the not-perfect world), the wealthy place a lower value on an extra dollar than the poor, and this poses problems when we use efficiency as a criterion for making decisions and deciding how resources are allocated. Some economists believe that economic efficiency and questions of how wealth is distributed should be dealt with separately, the former by economists and the latter by politicians. Politicians control the mechanisms for redistributing income, which are taxes, subsidies, and government ownership of certain industries. For our purposes, we make the simplistic assumption that government policy will lead to a fair distribution of wealth and so focus on economic efficiency.

If resource allocation is so easy using competitive markets, why is economics still practiced so widely? The answer is that markets do not always work and often fail to find an efficient allocation of resources. The causes of market failure are the next topic.

1.2.5 The Concept of Market Failure

A market will fail to find an efficient allocation of scarce resources for a number of reasons: market power, external costs and benefits, and the nature of certain goods and services.

Market Power: It is very important that no-one gains control of the market. Perfect competition relies on having many consumers and producers in the market. This ensures that every individual consumer and producer is a negligible part of the market and so has no power over how the market functions. If there are many small producers, it is hard for them to collude and increase the price above the natural market price. Producers must accept the market price. If there are many consumers then it is difficult for them to organize themselves to manipulate the price downwards. Also, every product traded must be identical in every way (i.e., homogenous). The rule of perfect competition encourages producers to compete on price alone and so minimize costs. Any deviation from perfect competition is likely to lead to an inefficient allocation of resources.

External costs and benefits: The supply of some goods and services in a market cause costs or benefits that impact on others and these are called externalities. The owners of a chemical plant who pollute the local water supply impose costs on others and the use of antibiotics may impose costs on others by causing resistant organisms to develop. These are both negative externalities. The converse are positive external benefits such as the occurrence of herd immunity arising from a vaccination program or the provision of a work-based keep fit program that encourages individuals to ride or walk to work and so parking becomes easier for those who have to drive. The presence of good and bad externalities will fool the market into an inefficient allocation of scarce resources.

The nature of some goods and services: Some goods and services are "public goods," which cannot be traded in perfectly competitive markets. Public goods are those that jointly benefit many people, yet it is difficult to exclude individuals from. An example is national defense. If scarce resources are allocated to defending a country, then the entire population is protected. It is impossible to select protection for some individuals and exclude others. For this reason, it is difficult to find out how individuals value the benefits of defense and without this information perfectly competitive markets cannot function. There are also goods and services that are not public goods that consumers still find difficult to value. For markets to work properly, consumers and producers must understand the quality, attributes, and prices in the market, so they know what they should buy to make themselves happy. Poor information may lead to mistakes and a failure to choose an efficient mix of goods and services. Information deficits can be remedied using an expert agent who has good knowledge of the product. An honest and hard working car mechanic will fulfill this role. This agency relationship can be problematic as the agent can induce the consumer to demand more of a good than they might have wanted if they had had perfect information. A dishonest car mechanic might convince you to refit a brake system that in reality will be perfectly safe for another 5,000 miles. This would not be an efficient outcome as you will have incurred positive cost and zero benefit.

Healthcare, including infection control, falls foul of most of these requirements [13]. It is very unlikely that we can use markets to allocate scarce healthcare resources efficiently; healthcare is an unusual good. For a start, no consumer actually wants healthcare, few people enjoy visiting hospital and being jabbed, poked, or chopped up. Lance Armstrong is a cancer survivor who has won the Tour de France many times. He made the following description of what it is like to demand healthcare:

> *"One thing they don't tell you about hospitals is how they violate you. It is like your body is no longer your own, it belongs to the nurses and doctors, and they are free to prod you and force things into your veins and various openings. The catheter was the worst; it ran up my leg into my groin, and having it put in and taken out again was agonizing. In a way, the small, normal procedures, were the most awful part of illness. At least for the brain surgery I'd been knocked out, but for everything else, I was fully awake, and there were bruises and scabs and needle marks all over me, in the backs of my hands, my arms, my groin. When I was awake the nurses ate me alive" [134].*

Consumers are unlikely to value the process of healthcare, but they do value the end product, improved health, and what that allows them to achieve in life, yet scarce resources are used to produce healthcare. There is a difference between health and healthcare, and the demand for healthcare services is derived from a demand for health. This is not a good start if we want to use perfectly competitive markets to decide how scarce resources are used in healthcare.

There are further problems. Producers often have market power in healthcare. Doctors tend to restrict entry to the medical profession by limiting the places available in medical schools and any healthcare professional must obtain the appropriate experience, qualifications, certifications, and licenses – for good reasons – before they enter the market. This means that consumers cannot always pick and choose who they purchase their goods and services from and healthcare producers often enjoy a monopoly; this is the opposite of perfect competition. Healthcare is not homogenous, everyone is slightly different and so producers can discriminate between consumers, robbing them of their collective bargaining power. With monopoly producers and many different products in the market, prices will tend to be higher than a competitive market would allow and this is not efficient. Healthcare markets are characterized by positive and negative externalities leading to further inefficiency in resource allocation and parts of health care are public goods, such as effective antibiotics. If we left the supply of antibiotics to the competitive market (as has happened in parts of south-east Asia), then the negative externality of antibiotic resistance would be ignored by the market, and this results is a poor allocation of resources. Consumers also lack information in health care markets. They will not know when they need healthcare, and after they get sick they will not know what's wrong with them, how to treat it, or whether they will prefer an aggressive treatment or some other approach. Instead, they rely on doctors and other healthcare professionals to express their demand for healthcare via an agency relationship. The advent of Health Maintenance Organizations (HMOs) in the United States offers more bargaining power over doctors on behalf of consumers, but the situation is still a long way from the perfect market.

It is unlikely that economic efficiency will be achieved via perfectly competitive markets in healthcare. Without the invisible hand to guide resource allocation, the costs and benefits of different decisions, and so the opportunity costs, must be measured manually. The process by which the costs and benefits of different uses of scarce healthcare resources is measured is called economic appraisal.

1.2.6 The Concept of Economic Appraisal

Economic appraisal is the manual collection of the information that competitive markets provide so effortlessly. We have seen that perfectly competitive markets gather all this information with apparent ease. Tim Harford, author of the Undercover Economist [19], describes the perfect market as:

> *"a giant supercomputer network. With amazing processing power and sensors in every part of the economy – reaching even inside our brains to read our desires – the market is constantly reoptimising production and allocating the results perfectly"*

Economic appraisal is what must happen when the supercomputer keeps crashing because of market failure. In the case of healthcare, the market will not work, and so the hard calculations about costs, benefits, and what is efficient to produce have to be made manually with a slide-rule and abacus. Economic appraisal is, therefore, used to help decision makers allocate scarce resources between competing alternatives when market mechanisms fail. This chapter began with two examples of economic appraisal. The evaluation of the costs and benefits of the "space program" vs. two other alternatives, and the evaluation of the costs and benefits of an "MRSA screening program" vs. two alternatives provided information about how resources might be allocated to improve efficiency.

One of the hardest parts of economic appraisal is to value the economic benefits of one activity over another. The competitive market requires consumers to reveal their valuations of the benefits of various goods and services relative to all other choices. When markets do not work, benefits have to be valued manually, and this is difficult. Attaching an economic value to improvements in health status and the avoidance of premature death is a real challenge and one that health economists have struggled with. A number of methods are used.

1. The Human Capital Approach: the projected future earnings of the individual are assumed to represent the economic value of life.
2. Socially Implied Valuations: the costs of public sector decisions that are designed to improve safety, such as road safety campaigns, imply the value of a human life.
3. Contingent Valuation: the value of healthcare programs are elicited from individuals who are asked to reveal the maximum they would be willing to pay to access some service or the minimum they would be willing to accept as a compensation for being denied access to a service.
4. Choice Experiments (or Conjoint Analysis): Individuals are asked to value a number of characteristics (or attributes) of the healthcare program such as waiting time, type of treatment, and type of staff who provided care. The valuations of these components can be used to identify the preference of valuation of a health program.

The "Contingent Valuation" and "Choice Experiment" methods are closest to how a market would find consumer valuations for goods and services, but these are still difficult to implement and have been criticized [20]. If the value of the benefits of healthcare can be accurately measured, then we are in a position to make some judgments about improving economic efficiency from reallocating resources in an economy, when markets will not do the job for us.

1.3 Conclusions

Making good choices over how scarce resources are allocated is important. Good choices provide more benefit to society than bad choices. Efficient resource allocation will maximize benefits, and governments should try to make sure all individuals get a fair slice of the wealth generated. Competitive markets allocate resources efficiently. When markets go wrong, as they do for health care, then decisions about how to allocate resources have to be made manually. The best way to inform those decisions is to conduct an economic appraisal. Economic appraisal is an important part of the health economists' toolkit and one that infection-control practitioners can readily exploit in order to move scarce resources toward infection control, or reorganize how existing infection-control resources are used. The infection-control practitioner can use economics to improve efficiency in health care resource allocation.